

IRINGA, Tanzania, United Republic of

WMO: 638870

Lat: 7.669S

Lon: 35.752E

Elev: 4678

StdP: 12.38

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	53.9	55.3	39.9	43.0	76.1	42.4	47.4	71.0	28.7	69.5	23.8	67.5	3.7	180	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	21.2	86.3	63.2	85.3	63.4	84.0	63.6	69.7	79.0	68.9	77.9	68.1	77.0	14.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.9	117.8	73.3	66.3	115.4	72.7	65.5	112.3	72.0	37.0	79.0	36.2	77.9	35.4	76.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.7	22.1	19.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	46.3	74.2	1.0	3.1	45.6	76.4	45.1	78.2	44.5	79.9	43.8	82.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	71.4	71.4	71.0	70.1	69.0	67.1	66.3	67.6	69.7	72.1	73.5	72.5
	DBStd	2.87	2.09	1.82	1.85	1.55	1.55	2.06	1.96	1.98	1.89	1.93	1.70	1.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	18	0	0	0	0	0	5	9	3	1	0	0	0
	CDD50	7354	664	600	651	603	589	514	507	545	591	685	706	699
	CDD65	1894	199	180	186	153	124	69	50	82	141	220	256	234
	CDH74	10906	843	895	862	569	501	361	337	501	1083	1744	1974	1236
	CDH80	1852	109	127	106	31	19	20	14	36	160	430	567	233
(14) Wind	WSAvg	6.4	3.7	3.8	3.2	5.1	7.0	8.0	8.6	7.9	8.1	8.8	8.2	4.8
(15) Precipitation	PrecAvg	24.7	5.2	4.2	5.0	2.6	0.4	0.0	0.0	0.0	0.0	0.2	1.3	5.8
	PrecMax	42.9	10.4	8.8	11.1	9.8	2.2	0.4	0.1	0.4	0.3	1.0	5.1	14.2
	PrecMin	18.3	0.8	1.2	1.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3
	PrecStd	5.5	2.3	1.9	2.3	2.1	0.5	0.1	0.0	0.1	0.1	0.3	1.2	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.3	85.7	84.6	82.9	82.2	82.4	82.0	83.3	86.2	88.0	88.2	86.5
		MCWB	65.7	65.4	68.0	66.2	63.7	62.5	60.8	61.7	61.5	61.1	63.1	64.8
	2%	DB	82.7	82.9	82.7	80.7	80.2	79.9	79.2	80.8	84.0	86.1	86.3	84.4
		MCWB	65.9	66.4	67.2	66.4	63.6	61.4	60.2	60.8	61.0	62.0	63.2	64.7
	5%	DB	80.8	81.4	80.9	79.0	78.7	77.9	77.4	78.9	82.1	84.4	85.3	82.7
		MCWB	66.5	66.5	67.4	66.1	63.2	60.8	59.7	59.8	60.8	61.9	63.1	64.7
	10%	DB	78.9	79.8	79.1	77.4	77.1	76.0	75.6	77.0	80.3	82.5	83.8	80.8
		MCWB	66.7	66.6	67.1	65.8	63.1	60.2	59.0	59.3	60.3	61.9	63.1	65.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	70.6	70.9	69.8	67.0	64.8	63.7	63.9	64.1	66.3	69.2	69.5
		MCDB	78.5	79.6	81.2	77.6	76.3	76.7	77.6	79.0	81.1	81.3	81.6	78.7
	2%	WB	69.0	69.2	69.4	68.3	65.4	63.3	62.0	62.0	63.0	64.8	67.3	68.2
		MCDB	77.3	78.2	79.1	76.8	76.1	76.5	77.3	78.1	79.9	80.5	79.0	76.7
	5%	WB	68.1	68.4	68.6	67.4	64.5	62.0	60.8	61.0	62.0	63.8	66.0	67.3
		MCDB	76.2	77.3	77.9	76.0	76.0	75.3	75.7	76.6	78.8	79.7	77.6	75.8
	10%	WB	67.3	67.5	67.5	66.6	63.8	61.0	59.6	60.0	61.1	63.0	65.1	66.6
		MCDB	75.2	76.1	76.4	75.2	74.9	74.1	73.8	75.1	77.8	78.9	77.3	75.2
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	17.4	17.4	16.5	17.8	19.9	20.6	20.7	22.5	22.2	21.2	17.7
		MCDBR	19.4	20.7	20.3	19.0	20.7	22.3	23.4	23.4	25.5	25.0	24.2	20.7
	5% WB	MCWBR	7.0	8.3	8.5	8.0	8.6	9.1	9.6	8.8	8.3	8.0	7.6	6.4
		MCWBR	16.2	17.6	18.2	16.5	18.1	21.2	22.8	21.8	23.2	22.6	20.3	16.7
(40) Clear-Sky Solar Irradiance	taub		0.385	0.380	0.353	0.314	0.313	0.310	0.326	0.357	0.362	0.371	0.349	0.361
		taud	2.435	2.458	2.551	2.753	2.646	2.570	2.469	2.358	2.351	2.336	2.451	2.490
	Ebn at Noon		303	303	306	310	302	298	295	292	299	302	312	309
		Edh at Noon	39	38	34	27	28	30	33	39	41	42	38	36
(44) All-Sky Solar Radiation	RadAvg		1751	1831	1795	1676	1615	1666	1715	1841	2060	2108	2022	1828
	RadStd		134	95	88	83	89	54	79	75	81	80	117	126

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page